Exercise No.3
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Data Analytics
BrightLearn
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```
20
21  --Question 1: A short product catalogue. Use the price column to bucket each item into a price tier.
22  select product_id,
23  product_name
24  price,
25  case
26   when price > 1000 then 'Expensive'
27   when price between 100 and 1000 then 'Mid-range'
28   when price < 100 then 'Budget'
29  end as price_category
30  from Products.sales.price;
31
32
```

Add parameter

Table +

	product_id	price	price_category
1	1	Laptop	Expensive
2	2	Phone	Mid-range
3	3	Keyboard	Budget
4	4	Monitor	Mid-range
5	5	Mouse	Budget

```
20
21  --Question 2:Customer orders with the amount paid. Label each order by its monetary tier.
22  select customer_name,
23  amount,
24  case
25  when amount >= 1000 then 'High Value'
26  when amount between 500 and 999.999 then 'Medium Value'
27  when amount <500 then 'Low Value'
28  end as order_value_category
29  from customers.spend.orders;
30
```

Add parameter

Table +

	customer_name	amount	order_value_category
1	Alice	150	Low Value
2	Bob	560	Medium Value
3	Charlie	999	Medium Value
4	Diana	45	Low Value
5	Ethan	1200	High Value

```sql
--Question 3: Combine the department and salary columns to label each employee with a seniority tier
  select emp_name,
  department,
  salary,
  case
  when department = 'IT' and salary > 80000 then 'Senior IT'
  when department = 'HR' and salary > 55000 then 'Experienced HR'
 else 'Staff'
 End as position_level
 from shop.employees.data;
```

Add parameter

Table

	emp_name	department	salary	position_level
1	John	IT	85000	Senior IT
2	Sara	HR	60000	Experienced HR
3	Mark	IT	75000	Staff
4	Lucy	Finance	950000	Staff
5	Tom	HR	55000	Staff

```sql
--Question 4:A list of student scores. Convert numeric scores into letter grades.
select student_name,
score,
case
when score >= 90 then 'A'
when score between 80 and 89 then 'B'
when score between 70 and 79 then 'C'
when score between 60 and 69 then 'D'
when score < 60 then 'F'
end as grades
from students.school.grades;
```

Add parameter

Table

	student_name	score	grades
1	Anna	92	A
2	Ben	76	C
3	Cara	59	F
4	David	83	B
5	Ella	68	D

```sql
--Question 5: Delivery times in minutes. Label each delivery by its performance.
select delivery_id,
delivery_time_minutes,
case
when delivery_time_minutes <= 30 then 'Fast'
when delivery_time_minutes between 31 and 60 then 'On Time'
when delivery_time_minutes > 60 then 'Late'
end as performance
from delivery.times.minutes;
```

Add parameter

Table

	delivery_id	delivery_time_minutes	performance
1	1	45	On Time
2	2	80	Late
3	3	30	Fast
4	4	65	Late
5	5	100	Late

```sql
--Question 6: Support tickets stored with a numeric priority. Convert the priority into a readable label.
select issue_type,
priority,
case
when priority = 3 then 'High'
when priority = 2 then 'Medium'
when priority = 1 then 'Low'
end as priority_label
from IT.issues.response;
```

Add parameter

Table

	issue_type	priority	priority_label
1	Login issue	1	Low
2	Server down	3	High
3	Slow system	2	Medium
4	Email error	2	Medium
5	Password reset	1	Low

```sql
--Question 7:Calculate each student's attendance percentage and classify the result.

select student_id,
(days_prsent/total_days)*100 as attendance_percentage
 from attendance.pupils.school;

 select student_id,
(days_prsent/total_days)*100 as attendance_percentage,
case
when attendance_percentage >= 90 then 'Excellent'
when attendance_percentage between 75 and 89 then 'Good'
when attendance_percentage <75 then 'Needs Improvement'
end as Attendance_Status
 from attendance.pupils.school;
```

Add parameter

Table

	student_id	attendance_percentage	Attendance_Status
1	1	90	Excellent
2	2	60	Needs Improvement
3	3	96	Excellent
4	4	50	Needs Improvement
5	5	100	Excellent

```sql
--Question 8: Stock levels per product. Label each row with the correct stock status
select product_id,
stock_qty,
case
when stock_qty = 0 then 'Out of Stock'
when stock_qty between 1 and 5 then 'Low Stock'
when stock_qty > 5 then 'In Stock'
end as stock_status
from products.inventory.stock;
```

Add parameter

Table

	product_id	stock_qty	stock_status
1	1	5	Low Stock
2	2	0	Out of Stock
3	3	25	In Stock
4	4	10	In Stock
5	5	3	Low Stock

```sql
--Question 9: Classes with their enrolled student counts. Classify each class by size

select subject,
enrolled_students,
case
when enrolled_students >= 25 then 'Large'
when enrolled_students between 10 and 24 then 'Medium'
when enrolled_students < 10 then 'Small'
end as class_size_category
from classes.subjects.students;
```

Add parameter

Table

	subject	enrolled_students	class_size_category
1	Math	30	Large
2	English	25	Large
3	Science	15	Medium
4	Art	5	Small
5	History	20	Medium

```sql
--Question 10:Combine the payment method and amount to decide whether a discount applies
select payment_id,
payment_method,
amount,
case
when payment_method = 'Cash' and amount >= 200 then 'Eligible for Discount'
else 'Not Eligible '
end as discount_eligibility
from payments.type.amount;
```

Add parameter

Table

	payment_id	payment_method	amount	discount_eligibility
1	1	Card	50	Not Eligible
2	2	Cash	200	Eligible for Discount
3	3	Card	150	Not Eligible
4	4	Paypal	75	Not Eligible
5	5	Cash	300	Eligible for Discount